

# Estimation of Quantum Fisher Information by variational quantum circuits on a quantum computer

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Supervisor: Prof. Enrico Prati

Co-supervisor: Dr. Sebastiano Corli

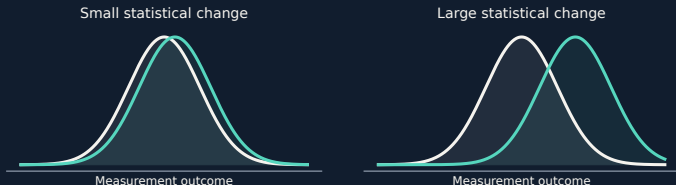
In collaboration with the **Italian Space Agency (ASI)**

Project: *Quantum Computers for Space Exploration*



# Classical Fisher Information

Local sensitivity, distinguishability of nearby parameter values.



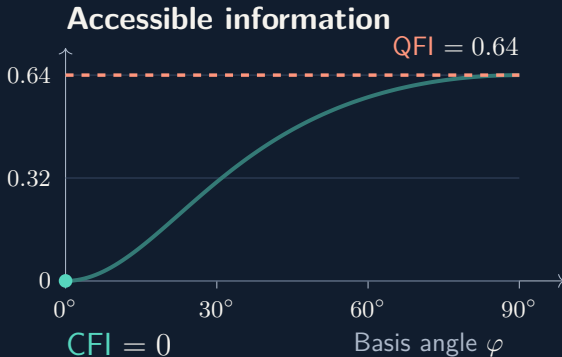
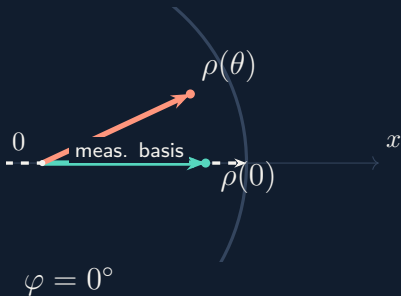
## Fisher Information

Sensitivity of an observed probability distribution to changes in an unknown parameter

How much information **a specific, fixed measurement** reveals about an unknown parameter governing a probability distribution.

# Fisher Information and Measurements

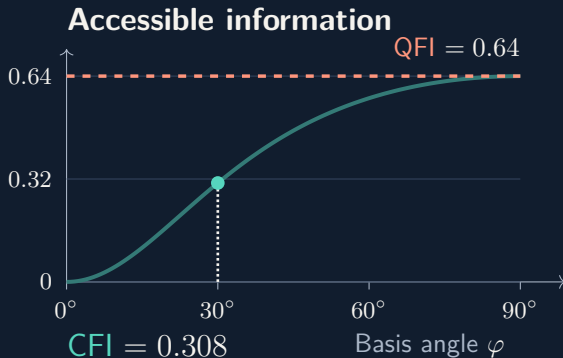
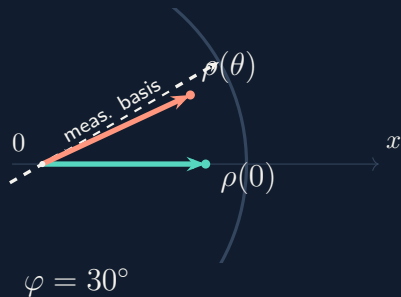
Two states inside the Bloch sphere



Mixed states,  $r = 0.8$ ; finite separation shown for clarity. CFI is local at  $\theta = 0$  (radians), not a two-state finite difference. Appendix J.

# Fisher Information and Measurements

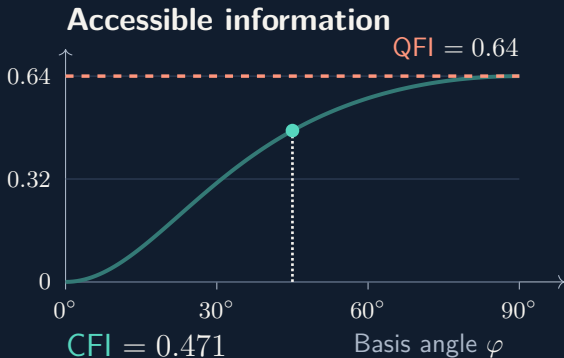
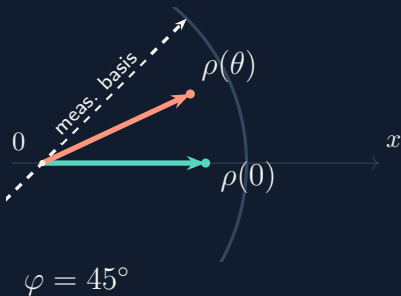
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# Fisher Information and Measurements

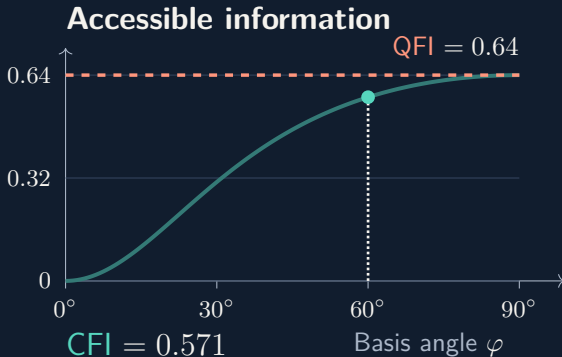
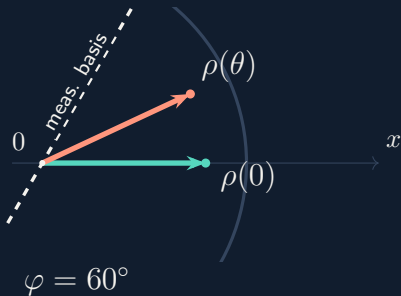
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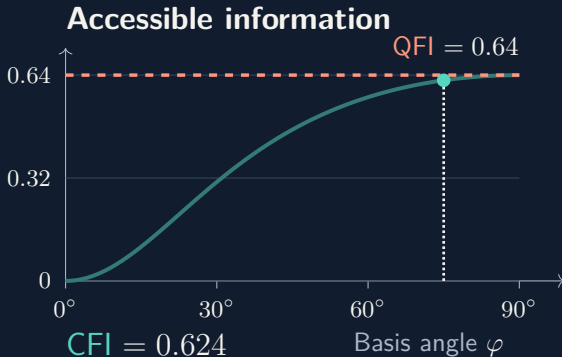
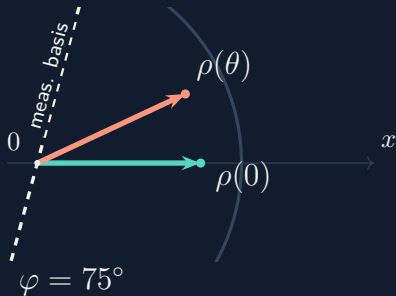
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# Fisher Information and Measurements

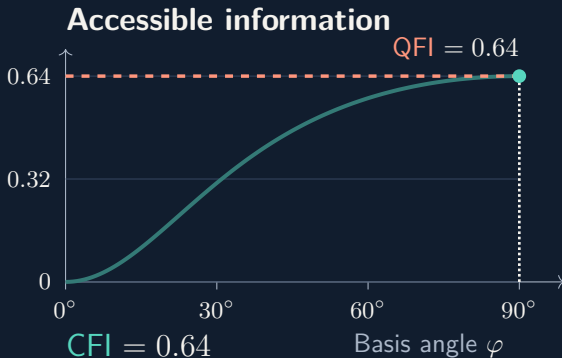
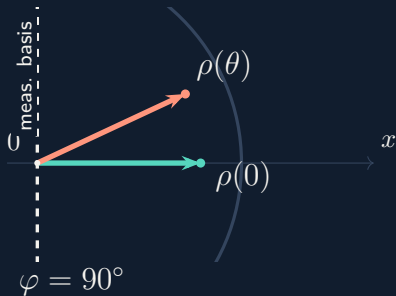
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# Fisher Information and Measurements

Two states inside the Bloch sphere



**QFI** is intrinsic to the quantum state  
and it is **independent of the measurement basis**

Mixed states,  $r = 0.8$ ; finite separation shown for clarity. CFI is local at  $\theta = 0$  (radians), not a two-state finite difference. Appendix J.

# How we deal with Mixed States

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**WHAT** Statistical mixtures of quantum states, described by a density matrix  $\rho$

Thesis: Introduction. Cerezo et al. (2022); Beckey et al. (2022). Subnormalized truncation; finite-step scope: Appendix A.

# How we deal with Mixed States

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**WHAT** Statistical mixtures of quantum states, described by a density matrix  $\rho$

**WHY** Real sensors are noisy and only partly accessible  
Mixed-state QFI remains less explored and costly to compute exactly

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# How we deal with Mixed States

**WHAT** Statistical mixtures of quantum states, described by a density matrix  $\rho$

**WHY** Real sensors are noisy and only partly accessible  
Mixed-state QFI remains less explored and costly to compute exactly

**HOW** Use truncated spectral decomposition

$$\rho_m = \sum_{i=1}^m \lambda_i |\psi_i\rangle \langle \psi_i|$$

The  $m$  largest eigenvalues and their eigenvectors.

## VQSE — Variational Quantum State Eigensolver

Learns the leading eigenvalues and a circuit that prepares their eigenvectors.

## TQFI — Truncated Quantum Fisher Information

Uses the retained components to construct fidelity-based QFI bounds.

Thesis: Introduction. Cerezo et al. (2022); Beckey et al. (2022). Subnormalized truncation; finite-step scope: Appendix A.

# Applications

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Study QFI and TQFI for two applications:

## Quantum magnetometry

Estimate a magnetic field with an interacting spin system.

## Exoplanet detection

Resolve a small exoplanet with spatial-mode measurements.

We used numerical simulation and quantum-circuits and sent ASI a complete report

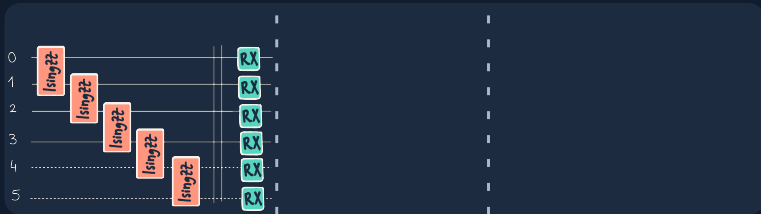
# Parallel numerical and quantum-circuit workflow

## NUMERICAL SIMULATION

### Preparation

Hamiltonian and  
time evolution

## QUANTUM CIRCUIT



Schematic circuit layouts and example output. Implementations can be exchanged at matched interfaces; physical stage order is preserved.

# Parallel numerical and quantum-circuit workflow

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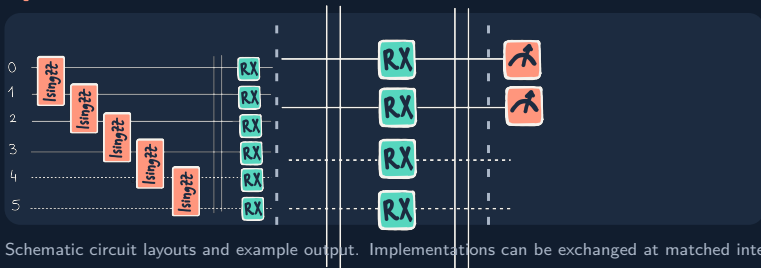
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Partial trace

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# Parallel numerical and quantum-circuit workflow

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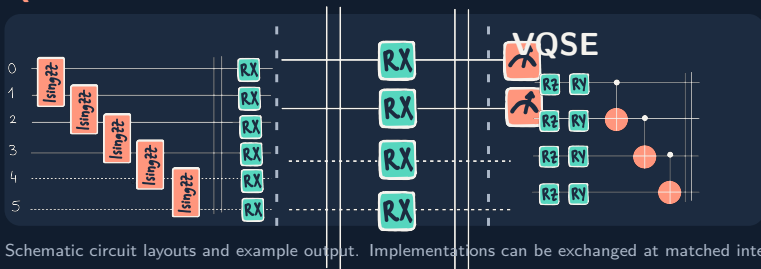
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### Spectrum

Exact spectral  
decomposition

## QUANTUM CIRCUIT



Schematic circuit layouts and example output. Implementations can be exchanged at matched interfaces; physical stage order is preserved.

# Parallel numerical and quantum-circuit workflow

## NUMERICAL SIMULATION

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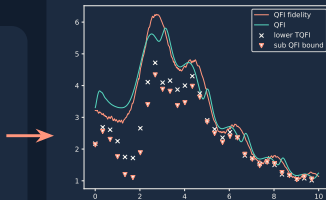
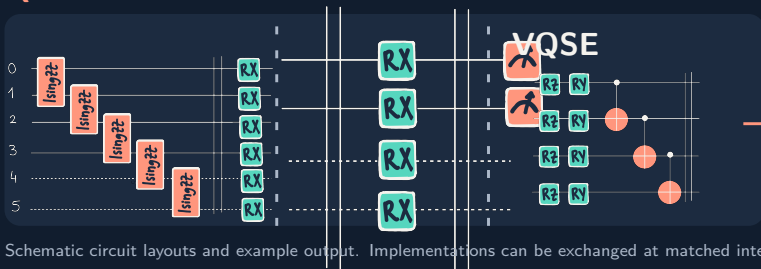
### Spectrum

Exact spectral  
decomposition

### Information

- 1) Exact SLD QFI
- 2) TQFI bounds

## QUANTUM CIRCUIT



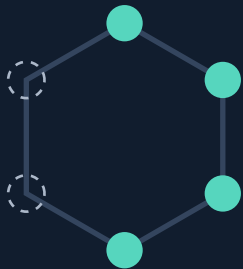
APPLICATION I

# Quantum Magnetometry

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# Toy Model: a noisy Quantum Magnetometer

Try multiple boundary conditions, preparations, noise models, to study QFI



Periodic spin chain

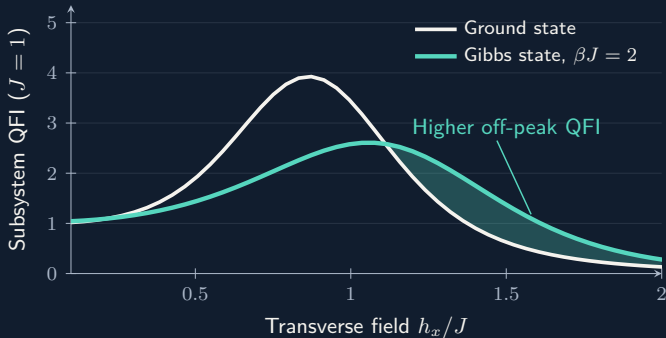
Partial Trace: 2 discarded

## Transverse-field Ising model (TFIM)

|              |                             |
|--------------|-----------------------------|
| Ground state | Lowest-energy preparation   |
| Thermal      | Field-dependent populations |
| Depolarized  | Fixed depolarizing noise    |

$N = 6$ ,  $n = 4$ , periodic boundaries,  $J = 1$ . Thesis: NoiseAnalysis.

# Some results: temperature broadens the range



## Broader sensitivity

More QFI beyond the ground-state peak.

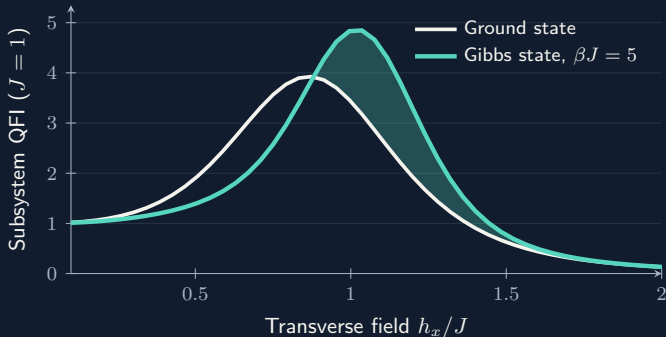
## Uncertain field

Useful across a wider field range.

**At  $\beta = 2$ , thermal preparation retains more information beyond the ground-state peak.**

Exact SLD QFI; periodic  $N = 6$ ,  $n = 4$ ,  $J = 1$ . Different preparation families. Thesis: NoiseAnalysis.

# Some more results: cooling strengthens peak



## Stronger peak

Higher QFI around the thermal optimum.

## Known field range

Useful near the target field.

**At  $\beta = 5$ , the thermal state exceeds the ground-state peak in the same spin subsystem.**

Exact SLD QFI; periodic  $N = 6$ ,  $n = 4$ ,  $J = 1$ . Different preparation families. Thesis: NoiseAnalysis.

APPLICATION II

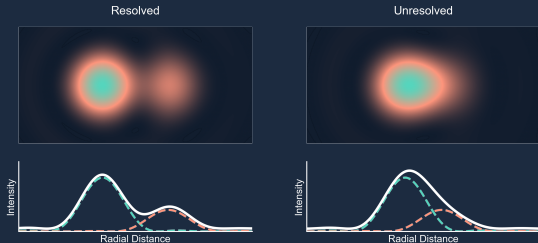
# Exoplanet Detection

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# The Exoplanet Problem

A faint exoplanet must be resolved in the presence of a bright star.

## Direct imaging



## Spatial-mode sorting



Photon counts in a spatial-mode basis.

**SPADE changes the measurement basis to reveal weak spatial signals.**

SPADE: SPAtial-mode DEMultiplexing. Tsang et al. (2016); Santamaria et al. (2024).

# SPADE modes and readout cross-talk

## $p_0$ : fundamental mode

Probability of a photon in  $HG_{00}$  dominated by the aligned star

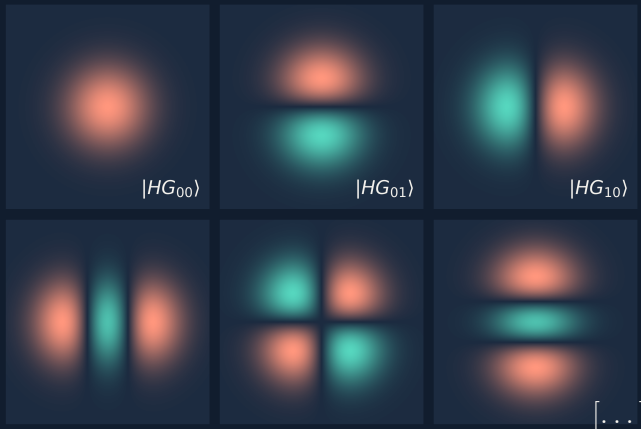
## $p_1$ : first-order modes

Probability in  $HG_{01}$  or  $HG_{10}$  sensitive to exoplanet separation

## $\chi$ : readout cross-talk

Probability of reporting port 1 instead of 0, or vice versa

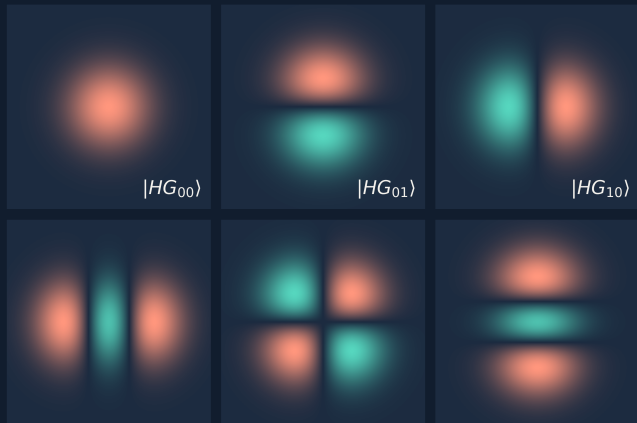
$$p_1^{\text{rep}} = \chi p_0 + (1 - \chi)p_1$$



Modal outcomes, not photon origin. Symmetric readout model; other outcomes unchanged. Thesis: Exoplanet Experiment.

# SPADE detector

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# Why describe the receiver with quantum channels?

The model includes mode populations and coherence before the measurement.

## CLASSICAL PROBABILITY EMBEDDING

$$\mathbf{p} = \begin{pmatrix} p_0 \\ p_1 \end{pmatrix} \mapsto \rho_{\text{diag}} = \begin{pmatrix} p_0 & 0 \\ 0 & p_1 \end{pmatrix}$$

Classical channel:  $\mathbf{p}' = R\mathbf{p}$

Redistributes probabilities between outcomes.

## QUANTUM COHERENCE

$$\rho = \begin{pmatrix} p_0 & c \\ c^* & p_1 \end{pmatrix}$$

Quantum channel:  $\rho' = \mathcal{E}(\rho)$

Acts on populations and coherences.

**Quantum channels describe optical effects before the detector produces classical counts.**

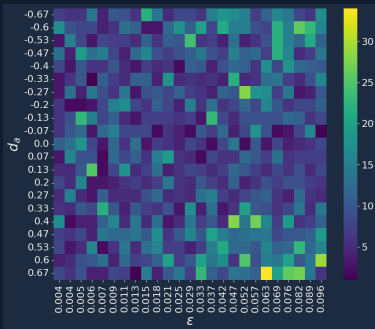
Two-outcome illustration:  $p_i \geq 0$ ,  $p_0 + p_1 = 1$ ,  $|c|^2 \leq p_0 p_1$ . A state vector of probabilities does not determine a quantum channel.

# Experimental data

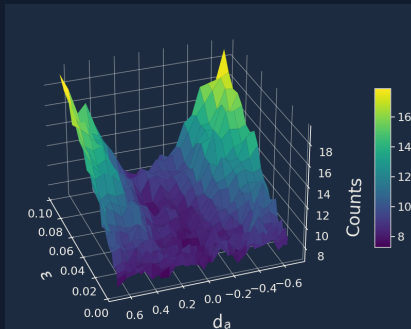
The completed ASI Matera experiment, with separation  $d$  and relative intensity  $\epsilon$ .

52,500 count observations: 525 settings  $\times$  100 repetitions; 10 ms each

ONE REPETITION



MEAN OF 100



## Missing observables

Separate modal outputs

Other measurement bases

Incident flux and failure outcomes

# From physical assumptions to predicted counts

A modular model connects source preparation, optical effects and detector readout.



Choose physically motivated channels and check their mathematical validity.

## Physical channel choices

Kraus operators for loss, dephasing and coherent mode mixing.

## Classical calibration

Fit throughput, background, offset and coupling from recorded counts.

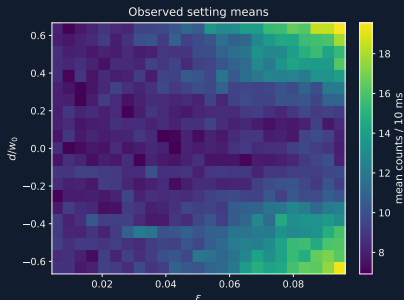
**Simulate the chosen channel sequence and compare its output with the observations.**

Thesis: provisional channel study. Optical assumptions are replaceable; the count record does not determine their individual values. Details: Appendices K–M.

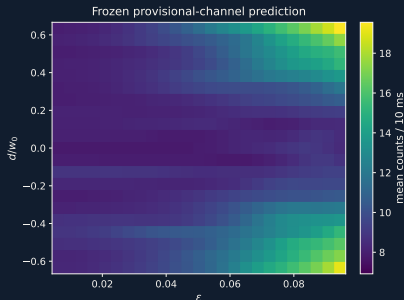
# The model reproduces the observed mean response

Both maps show a central minimum and larger counts at greater separation and source ratio.

## Observed setting means



## Modelled setting means



**The mean-count pattern is captured by the chosen channels with classical calibration.**

Thesis appendix: provisional channel model, top panels. In-sample mean comparison; assumed optical channels are not individually identified by this agreement.

# Conclusions

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Simulation, preparation studies, data interpretation and a variational optical proposal.

- **Perfectly matching quantum and numerical sensing pipeline**

Simulated quantum sensors under noise and decoherence, estimating QFI and TQFI.

- **Preparation and noise models**

Compared ground and Thermal states, depolarization and restricted access to the sensor.

- **Quantum-channel interpretation of SPADE data**

Modelled SPADE with quantum channels and fitted the observed mean counts.

- **VQ-SPADE: a proposed optical eigensolver**

A trainable mode rotation and SPADE readout would diagonalize the optical state.

Thesis: Methods, Noise Analysis, Exoplanet Experiment and Conclusions. VQ-SPADE is proposed for a future experiment.

# Spectral truncation and the local-QFI limit

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Dominant eigencomponents bound a fidelity quantity at finite parameter displacement.

$$\mathcal{I}_{\Delta} = \frac{8[1 - F(\rho_{\theta}, \rho_{\theta+\Delta})]}{\Delta^2}$$

$F$  is root fidelity.

The retained eigenspace is state dependent; it is not a fixed detector-mode cutoff.

At regular operating points,  $\mathcal{I}_{\Delta} \rightarrow \mathcal{F}_Q$  as  $\Delta \rightarrow 0$ .

**Finite-step bounds need not be pointwise bounds on the exact SLD QFI.**

Thesis: Methods; Mathematical Background. Cerezo et al. (2022); Beckey et al. (2022).

# The ansatz metric and its scope

---

The physical sensing parameter and the trainable circuit controls define different derivatives.

$$F_{\mu\nu}^{\text{ans}} = 4 \operatorname{Re}[\langle \partial_\mu \psi | \partial_\nu \psi \rangle - \langle \partial_\mu \psi | \psi \rangle \langle \psi | \partial_\nu \psi \rangle]$$

Pure ansatz:  $F^{\text{ans}} = 4g_{\text{FS}}$ , the Fubini–Study metric up to convention.

$$\min_{\Delta\alpha} \nabla C^\top \Delta\alpha + \frac{1}{2\eta} \Delta\alpha^\top (F^{\text{ans}} + \lambda I) \Delta\alpha$$

Stationarity gives the regularized natural-gradient update.

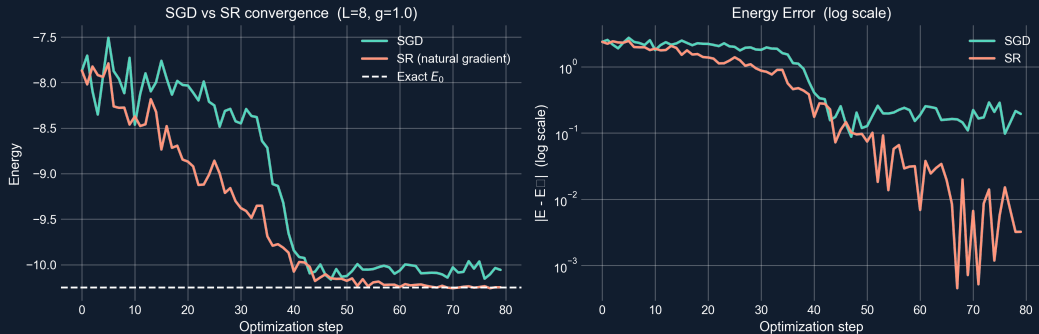
The quadratic term constrains state change, not objective curvature.

**Mixed VQSE inputs require a mixed-state metric; a pure-state metric is otherwise a surrogate.**

Thesis: Mathematical Background; VQSE. A full metric has quadratic parameter-count overhead; no general training speedup is claimed.

# Stochastic reconfiguration in a neural quantum state

A separate pure-state RBM example illustrates metric-based optimization.



**SR lowers the energy error in this run; it does not benchmark mixed-state VQSE.**

Thesis figure: VQSE chapter. TFIM at  $\hbar_x/J = 1$ ; 80 steps, 300 Monte Carlo samples/step,  $\eta = 0.05$ , SR  $\lambda = 10^{-4}$ ; same seed.

# Thermal sensitivity and data processing

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The Thermal and ground-state families need not have the same field response.

$$\rho_{\beta}(h_x) = \frac{e^{-\beta H(h_x)}}{\text{Tr } e^{-\beta H(h_x)}}$$

The field changes both eigenvectors and thermal populations.

Partial trace still obeys data processing within each family:

$$\mathcal{F}_Q[\text{Tr}_E \rho_{\beta}(h_x)] \leq \mathcal{F}_Q[\rho_{\beta}(h_x)]$$

For an uncertain field, choose an operating range or prior.

These curves do not establish a universal Bayesian or minimax optimum.

**The comparison changes the preparation family; it does not reverse a fixed noise channel.**

Thesis: Noise Analysis. Finite system; exact SLD QFI.

# Proof of Kraus-unitary freedom

---

A change of Kraus basis leaves the channel unchanged.

$$\widetilde{K}_a = \sum_b u_{ab} K_b, \quad U^\dagger U = I$$

$$\sum_a u_{ab} u_{ac}^* = \delta_{bc}$$

Orthonormal columns of  $U$ .

$$\sum_a \widetilde{K}_a \rho \widetilde{K}_a^\dagger = \sum_{b,c} \delta_{bc} K_b \rho K_c^\dagger = \sum_b K_b \rho K_b^\dagger$$

Isometry in the Kraus index leaves the action on every input unchanged.

**A physical realization adds assumptions about the environment; the reduced map alone does not select them.**

Standard Kraus equivalence: an isometry, or a unitary after padding with zero operators. Algebra shown explicitly for this presentation.

# A classical response does not fix a quantum channel

---

Embedding a stochastic map is a model construction, not process tomography.

$$\mathbf{p} \mapsto \text{diag}(\mathbf{p}), \quad \mathbf{p}' = R\mathbf{p}$$

For a column-stochastic  $R$ , one valid quantum extension is

$$K_{ji} = \sqrt{R_{ji}}|j\rangle\langle i|, \quad \mathcal{E}_R(\rho) = \sum_{ji} R_{ji} \rho_{ii} |j\rangle\langle j|$$

This extension measures and prepares in the chosen basis, erasing coherences.

For  $R = I$ , both the identity channel and complete dephasing reproduce the same populations.

**Different channels can agree on the measured populations; this is distinct from Kraus freedom of one channel.**

Thesis: classical/quantum embedding and HG coherence discussion. Channel-extension example derived for the presentation.

# The loss map and its physical checks

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The input mode space is embedded in an output space containing an orthogonal failure flag.

$$\mathcal{L}_\tau(\rho) = K_t \rho K_t^\dagger + \text{Tr}[(I - K_t^\dagger K_t)\rho] |\text{fail}\rangle \langle \text{fail}|$$

For any ancillary system, Kraus form preserves positivity:

$$(\mathcal{L} \otimes \text{id})(X) = \sum_a (K_a \otimes I) X (K_a^\dagger \otimes I) \geq 0 \text{ for } X \geq 0.$$

Boundary checks:  $\tau_j = 1$  transmits mode  $j$ ;  $\tau_j = 0$  sends it to failure.

Normalization: transmission plus failure has unit trace.

Physical validation: mode-injection measurements determine the  $\tau_j$ .

**Renormalizing the transmitted branch alone changes the question to postselected performance.**

Thesis: mode-dependent loss and flagged mode reduction. The record lacks independent modal-transmission calibration.

# Predictive calibration of the observed response

A physical mean with a Gaussian-process correction predicts held-out setting means.

$$\mu(d_a, \epsilon) = \mu_0(d_a, \epsilon) \exp[f(d_a, \epsilon) + \xi]$$

Grouped five-fold validation; all components refitted within each training fold.

| Held-out structure   | RMSE  | Standardized SD | 95% coverage |
|----------------------|-------|-----------------|--------------|
| Checkerboard cells   | 0.606 | 1.004           | 93.52%       |
| Distance rows        | 0.734 | 1.041           | 94.67%       |
| Source-ratio columns | 0.609 | 1.009           | 93.71%       |

**The validated target is the mean of 100 repetitions; this does not identify a complete quantum channel.**

Thesis: Exoplanet Experiment Appendix, hierarchical calibration validation. RMSE in counts per 10 ms.

# From SPADE to a variational optical measurement

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VQ-SPADE is a proposed optical realization of variational state eigensolving.

$$p_k(\boldsymbol{\alpha}; d_a) = \langle HG_k | V(\boldsymbol{\alpha}) \rho(d_a) V^\dagger(\boldsymbol{\alpha}) | HG_k \rangle$$

A programmable coherent mode transformation provides the trainable basis.

Resolved outputs, flux and failure events provide normalized probabilities.

Nearby states and additional overlap measurements support fidelity-based QFI bounds.

**The present record motivates the acquisition design; it does not implement this protocol.**

Thesis: proposed VQ-SPADE scheme. Additional independent channel and background calibrations are required.

# Mixed states and measurement-dependent information

---

Two finite-separated vectors illustrate one state family; Fisher information is evaluated locally.

$$\rho(\theta) = \frac{I + r(\cos \theta X + \sin \theta Z)}{2}, \quad r = 0.8, \quad \rho \equiv \rho(0)$$

$$M_{\pm}^{(\varphi)} = \frac{I \pm (\cos \varphi X + \sin \varphi Z)}{2}, \quad p_{\pm}(\theta) = \frac{1 \pm r \cos(\theta - \varphi)}{2}$$

$$\mathcal{F}_C(\varphi)|_{\theta=0} = \frac{r^2 \sin^2 \varphi}{1 - r^2 \cos^2 \varphi} \leq r^2 = \mathcal{F}_Q$$

At  $\varphi = 0^\circ$ : CFI = 0. At  $45^\circ$ : CFI =  $8/17 \simeq 0.471$ .

At  $\varphi = 90^\circ$ , measuring  $Z$  attains QFI = 0.64; the SLD is  $L_0 = rZ$ .

Both displayed vectors have radius 0.8; their  $25^\circ$  separation is only illustrative.

**Changing the measurement changes accessible information, not the underlying quantum limit.**

Exact mixed-qubit model;  $\theta$  in radians. Full rank keeps every displayed measurement regular. No finite-difference QFI approximation is used.

# Kraus representations are not unique

Different operator sets can describe exactly the same quantum channel.

$$\mathcal{E}(\rho) = \sum_a K_a \rho K_a^\dagger, \quad \sum_a K_a^\dagger K_a = I$$

Example: complete dephasing of a two-mode state in the chosen basis.

## Basis projectors

$$K_0 = |0\rangle\langle 0|, \quad K_1 = |1\rangle\langle 1|$$

## Random phase flips

$$\widetilde{K}_0 = \frac{I}{\sqrt{2}}, \quad \widetilde{K}_1 = \frac{Z}{\sqrt{2}}$$

$$\sum_a K_a \rho K_a^\dagger = \frac{\rho + Z\rho Z}{2} = \text{diag}(\rho)$$

**Kraus freedom changes the representation. Physical hypotheses concern the channel.**

Two-mode illustration of the thesis HG-dephasing model at  $v = 0$ . Equivalent representations cannot be distinguished by system-only data.

# A physical construction: mode-dependent loss

Mode transmission  $0 \leq \tau_j \leq 1$ ; loss is represented by an orthogonal failure state.

**Choose operators with explicit physical roles**

$$K_t = \sum_j \sqrt{\tau_j} |j\rangle \langle j|, \quad K_j^f = \sqrt{1 - \tau_j} |\text{fail}\rangle \langle j|$$

Transmission and failure jointly account for every input photon.

$$K_t^\dagger K_t + \sum_j (K_j^f)^\dagger K_j^f = \sum_j [\tau_j + (1 - \tau_j)] |j\rangle \langle j| = I$$

Kraus form ensures complete positivity; completeness proves trace preservation.

**Channel validity can be proved mathematically; physical suitability requires calibration.**

Derived from the thesis mode-dependent loss map. Input: one-photon mode space after common survival; failure is retained in the output.

# Information retained by the optical receiver

Compare quantum and classical information at successive stages of the receiver.

$$\mathcal{F}_Q(\rho_S) \geq \mathcal{F}_Q[\mathcal{E}(\rho_S)] \geq \mathcal{F}_C(\mathbf{p}) \geq \mathcal{F}_C(R\mathbf{p})$$

For channels and measurements independent of the parameter being estimated.

## Checkpoint

## Fraction of source QFI

|                                                       |            |
|-------------------------------------------------------|------------|
| ● Resolved reference, first-order and failure outputs | 99.68–100% |
| ● Binary readout; assumed 0.35% port confusion        | 0.13–57.0% |

**Aggregation and readout confusion can strongly reduce the accessible information.**

Conditional model: same nine settings, per photon surviving common loss. Ranges across settings; not reconstructed experimental QFI.